Derivative Securities FNCE 30007

Tutorial 1

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History of Forwards

Historical Example: U.S. Wheat Farmers in the mid-19th century

- Suppose a farmer plants grains in spring and expects to harvest in summer. He might worry about the price of grain in summer.
  - ”What if there is a way to preemptively know how much I can sell at?”
- On the other hand, a merchant might also think the weather condition is not that good in the recent days, so he might also worry about the uncertain price of grain.
  - ”What if there is a way to lock in the price that I will buy in the summer?”
- To lock in a stable price, a *forward contract* was agreed: “I sell 5,000 bushels in June at $X per bushel, agreed today.”
- This commitment securitized harvest revenue, giving both parties certainty.
- This is a **forward-like** commitment.

Review

A futures contract is an agreement to buy or sell an asset at a certain time in the future for a certain price.

- The futures price is the price at which you agree to buy or sell.

A long (short) futures position is an agreement to buy (sell) the asset at a certain time in the future for a certain price.

- No exchange of the underlying asset or cash occurs at contract signing.

Review

What does the contract specify?

- It specifies what can be delivered (i.e. the acceptable asset quality or grade)
    - Settlement may be either physical delivery or cash settled. Physical delivery means transferring the actual underlying asset; cash settlement means only settling the price difference.
- It specifies where delivery must take place (approved delivery locations or warehouses) and any price adjustments for alternative locations.
- It specifies when delivery may occur — usually defined delivery months (e.g. March, June, September, December).
- It defines contract size (e.g. 5,000 bushels of corn, 1,000 barrels of crude oil, etc.).

Usually, the party with the short position can choose the delivery date.

Review

Why use derivative securities?

- Hedge: Example: firms does not want unpredictable movement of its product (or product input) price.
- Speculate: Betting on his/her view (bullish/bearish) to earn money.
- Arbitrage: Earn risk-free profit almost for sure.

Review

Close out position

- Closing out a position involves entering into an opposite trade to the original one that opened the position.
- Example: Question 5.

Review

Clearinghouse and Margin Account

- A clearing house acts as an intermediary in futures transactions. It guarantees the performance of the parties to each transaction.
- A margin is cash (or marketable securities) deposited by an investor with his or her broker.
  - Brokers are required to maintain margin accounts with clearinghouse members, and brokers are required to maintain a margin account with the clearinghouse.
- The balance in the margin account is adjusted to reflect daily settlement.
- Margins minimize the possibility of a loss through a default on a contract.
- Three concepts: initial margin; maintenance margin; variation margin.

Review

This example illustrates how daily settlement (mark-to-market) works: each day the futures price changes, the contract is re-valued and the margin account is adjusted accordingly:

An investor takes a long position in one December 2010 ASX 200 futures contract on 20 October. (Note that 1 index points equals to \$25) The futures price is 4,636 index point, the initial margin requirement: AUD 3,250 per contract, maintenance margin is AUD 3,150. The fluctuation in the futures price will incur daily settlement, the details is in the next slide.

- Note that the contract size: AUD 115,900 $(= 4{,}636 \times 25)$

Review

Day	Futures Price (index points)	Daily Gain/Loss (AUD)	Cumulative Gain/Loss (AUD)	Margin Account Balance (AUD)	Margin Call (AUD)
	4,636.0			3,250	
20 Oct	4,632.0	–100	–100	3,150	—
21 Oct	4,630.0	–50	–150	3,100	150
22 Oct	4,649.0	475	325	3,725	—
25 Oct	4,719.0	1,750	2,075	5,475	—
26 Oct	4,689.0	–750	1,325	4,725	—
27 Oct	4,643.0	–1,150	175	3,575	—

- When the margin account falls below the maintenance margin (AUD 3,150)—as it does on 21 October—a margin call is triggered. Here, the investor adds AUD 150 (variation margin) to restore the balance back to the initial margin level. This ensures continued contract validity.

Tutorial Question

Q1: The part with a short position in a futures contract sometimes has options as to the precise asset that will be delivered, where delivery will take place, when delivery will take place, and so on. Do these options increase or decrease the futures price? Explain your reasoning.

- What is the incentive for the short side given they have some degree of freedom in making such decisions?
  - ”maybe deliver to the closest warehouse rather than directly to the home?”
  - ”maybe deliver by the end of deadline rather than earlier?”

These options make the contract less attractive to the long position and more attractive to the short position. This tends to reduce the futures price.

Tutorial Question

Q2: Explain how margins protect investors against the possibility of default.

- Default risk/ Settlement risk.
- Margin is money deposited by a trader with his or her broker, by the broker with the clearing house member, and by the clearing house member with the clearing house. It acts as a guarantee that any losses on the futures contract will be covered. The balance in the margin account is adjusted daily to reflect gains and losses on the futures contract. If losses reduce the balance to below a certain level (the maintenance margin), a trader is required to deposit further margin with the broker. This system makes it less likely that the trader will default.

Tutorial Question

Q3: An investor enters into two long July futures contracts on frozen orange juice. Each contract is for the delivery of 15,000 pounds. The current futures price is 160 cents per pound, the initial margin is $6,000 per contract, and the maintenance margin is $4,500 per contract. What price change would lead to a margin call? Under what circumstances could $2,000 be withdrawn from the margin account by the investor?

- There is a margin call if more than $6,000 (IM) - $4,500 (MM) =$1,500 is lost on one contract.
  - Then how much futures price fall will translate to $1500 lost on one contract?
  - 15000 pounds * x cent/pound = $1500, $x = 10$
  - This happens if the futures price of frozen orange juice falls by more than 10 cents to below 150 cents per lb.
- $2,000 can be withdrawn from the margin account if there is a gain on one contract of $1,000. This will happen if the futures price rises by 6.67 cents to 166.67 cents per lb.

Tutorial Question

Q4: Show that, if the futures price of a commodity is greater than the spot price during the delivery period, there is an arbitrage opportunity. Does an arbitrage opportunity exist if the futures price is less than the spot price? Explain your answer.
If the futures price greater than spot price i.e. $F_T > S_T$ at maturity,

- Arbitrageur buys the asset, shorts the futures and makes delivery for immediate profit.

If the futures price less than spot price i.e. $F_T < S_T$ at maturity, there is no similar perfect arbitrage.

- The arbitrageur can take a long futures position but cannot force immediate delivery of the asset. The decision on when delivery will occur is made by the short party.
- One day after, it is possible that the spot price drops a lot...

Tutorial Question

Q5: On July 1, 2007, a U.S. company enters into a forward contract to buy 10 million British pounds on January 1, 2008. On September 1, 2007, it enters into a forward contract to sell 10 million British pounds on January 1, 2008. Describe the profit or loss the company will make in dollars as a function of the forward exchange rates on July 1, 2007, and September 1, 2007.

- Let F_1 and F_2 denote the forward exchange rates for contracts entered into on July 1, 2007 and September 1, 2007, respectively, and let S_T denote the spot rate on January 1, 2008. All exchange rates are measured as dollars per pound.

Tutorial Question

Q5:

- Payoff from the first contract:

$$10 \times (S_T - F_1) \text{ million dollars}$$

- Payoff from the second contract:

$$10 \times (F_2 - S_T) \text{ million dollars}$$

- The total payoff is therefore

$$10 \times (S_T - F_1) \; + \; 10 \times (F_2 - S_T) \; = \; 10 \times (F_2 - F_1) \text{ million dollars.}$$

- Opening and closing position.

Tutorial Question

Q6: What do you think would happen if an exchange started trading a contract in which the quality of the underlying asset was incompletely specified?

- The contract would not be a success. Short parties would deliver the cheapest form of the asset which may be totally unacceptable to the long party. Once the quality problem became well known no-one would be prepared to go long. Futures contracts are only feasible when there are rigorous standards defining the quality of the asset. Many futures contracts have failed because of the difficulties associated with defining quality.

Tutorial Question

Q7:An airline executive has argued: 'There is no point in our using oil futures. There is just as much chance that the price of oil in the future will be less than the futures price as there is that it will be greater than this price.' Discuss the executive's viewpoint.

- It may well be true that there is just as much chance that the price of oil in the future will be above the futures price as that it will be below the futures price. This means that the use of a futures contract for speculation would be like betting on whether a coin comes up heads or tails. But it might make sense for the airline to use futures for hedging rather than speculation. The futures contract then has the effect of reducing risks.